A PROJECT REPORT ON

PREPAI: AN AGENTIC VOICE-DRIVEN TECHNICAL INTERVIEW PLATFORM WITH ADAPTIVE FOLLOW-UP AND RESUME RETRIEVAL-AUGMENTED GENERATION

Submitted in partial fulfillment of the requirements
for the award of the degree of

BACHELOR OF TECHNOLOGY
in
COMPUTER SCIENCE AND ENGINEERING

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PROJECT TITLE PAGE

PREPAI: AN AGENTIC VOICE-DRIVEN TECHNICAL INTERVIEW PLATFORM WITH ADAPTIVE FOLLOW-UP AND RESUME RETRIEVAL-AUGMENTED GENERATION

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Academic Session: 2025-2026

CERTIFICATE OF APPROVAL

This is to certify that the project entitled "PrepAI: An Agentic Voice-Driven Technical Interview Platform with Adaptive Follow-Up and Resume Retrieval-Augmented Generation" submitted by Ansh Verma (Roll No. 210103010042) in partial fulfillment of the requirements for the award of the Degree of Bachelor of Technology in Computer Science and Engineering is an authentic record of bonafide research and developmental work carried out under my personal supervision and guidance during the academic session 2025-2026.

The results and implementations embodied in this project report have been verified through automated functional tests, empirical speech transcription and inference benchmarks, connection pool validation with Locust, and interview rubric scoring calibration. The content has not been submitted in part or in full to any other University or Institute for the award of any other degree, diploma, or fellowship.

Date: 14th October 2026
Place: Department of Computer Science & Engineering

Project Supervisor / Guide Dr. R. K. Sharma, Ph.D. Professor, Dept. of CSE	Head of Department Dr. S. Mukherjee, Ph.D. Professor & Head, Dept. of CSE
Internal Examiner	External Examiner

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ABSTRACT

Technical job interviews in modern software engineering demand clear verbal articulation, conceptual depth, and the ability to defend architectural trade-offs under scrutiny. However, aspiring software engineers frequently lack access to realistic interview practice. Human mock interview coaching is financially prohibitive ($100-$300/hour), static question listings fail to train spoken communication, and generic LLM chatbots operate only via text without resume grounding or dynamic follow-up questioning.

This project presents PrepAI, an open, full-stack, voice-driven mock interview platform powered by Agentic AI and Retrieval-Augmented Generation (RAG). Unlike conventional one-turn question-and-answer systems, PrepAI introduces an autonomous Adaptive Follow-Up Agent. When a candidate verbally answers a technical question, the agent evaluates response depth in real time. If the answer is superficial or misses key trade-offs, the agent dynamically generates a targeted probe question to test deeper comprehension. Once sufficient depth is demonstrated or maximum probe depth is reached, it concludes the turn with a balanced recruiter critique, rubric score (1-10), and a Staff-Engineer level model answer.

The platform combines a browser-native voice recording pipeline (HTML5 MediaRecorder) with hardware-accelerated speech-to-text via Groq-hosted Whisper-large-v3-turbo, achieving speech transcription in approximately 380-500 ms. Candidate resumes are processed through a dynamic RAG pipeline: PDF text is split into overlapping sliding windows, embedded into 384-dimensional dense vectors via Cohere API, indexed natively in PostgreSQL using pgvector, and automatically cleaned up upon resume re-upload to eliminate stale context.

To guarantee backend stability during multi-user sessions, relational operations are routed through transaction-level connection pooling (Supabase PgBouncer, port 6543) with SQLAlchemy pre-ping verification, validated via Locust load tests simulating 50 to 100 concurrent active users. In addition to interview simulation, PrepAI provides career tooling including an ATS job description matcher, tailored cover letter generator, and LinkedIn profile bio synthesizer. Experimental evaluations

on 30 benchmark interview responses demonstrate an average scoring Mean Absolute Error (MAE) of 0.42 points compared to senior engineering recruiter rubrics, confirming PrepAI as an effective, realistic, and responsive technical interview preparation system.

Keywords: *Agentic AI, Adaptive Follow-Up, Retrieval-Augmented Generation (RAG), Speech-to-Text Transcription, PostgreSQL pgvector, Groq Cloud, Connection Pooling, Technical Interview Simulation.*

CHAPTER 1: INTRODUCTION

1.1 Project Overview & Background

Technical hiring in modern software engineering centers on rigorous, verbal interviews. Hiring managers and senior engineers evaluate candidates not simply on rote code syntax, but on verbal communication, architectural reasoning, trade-off analysis, and past project execution. During interviews, candidates are expected to verbally explain concurrency models, defend database indexing strategies, articulate debugging journeys, and break down system bottlenecks under time constraints.

Despite the critical importance of interview performance, access to high-quality technical practice remains limited. Professional mock interview coaching costs between $100 and $300 per hour, making frequent practice impractical for university students and early-career engineers. Static question websites (such as LeetCode discussion forums) offer passive reading without vocal articulation. While general conversational AI models (such as basic chat interfaces) have emerged, they operate primarily through typed text, lack grounding in the candidate's actual resume projects, and fail to simulate the interactive probing behavior of real human interviewers.

PrepAI was developed to provide an open, voice-driven, interactive technical interview simulator. By combining fast speech-to-text processing, resume Retrieval-Augmented Generation (RAG), and an autonomous Adaptive Follow-Up Agent, PrepAI delivers realistic interview simulations tailored to each candidate's experience.

1.2 Evolution of Automated Interview Systems

Automated interview practice tools have evolved through three major stages:

1. Rule-Based & Keyword Systems (First Generation): Early platforms matched candidate answers against fixed keyword patterns and regex rules. These systems lacked semantic understanding, penalized alternative phrasing, and could not hold natural conversations.

2. Generic LLM Chatbots (Second Generation): With the rise of foundational LLMs, developers built simple prompt-based chatbots. While fluent, these tools suffered from high conversational latency (4-8 seconds), hallucinations, and static

single-turn questioning where every answer was simply assigned a score without probing.

3. Agentic Voice & RAG Platforms (Third Generation - PrepAI): PrepAI advances beyond static chatbots by introducing an autonomous Adaptive Follow-Up Agent. It combines fast Whisper speech transcription on Groq hardware, resume-grounded RAG via PostgreSQL pgvector, multi-turn probe interrogation, and audio dictation controls.

1.3 Key Innovations & Significance

The engineering contributions of PrepAI include:

• Autonomous Adaptive Follow-Up Agent: When a candidate provides an initial answer, the agent dynamically evaluates its depth. If the response is superficial, it generates an intelligent follow-up probe; if senior depth is shown, it finalizes the score and provides a Staff-level model answer.

• Direct LLM Orchestration without Heavy Frameworks: The agentic loop is implemented directly using Python asyncio and Groq API calls without the overhead or unpredictability of heavy graph libraries like LangGraph.

• Resume Grounding via PostgreSQL pgvector: Candidate project experiences are extracted, embedded via Cohere API, and queried using native cosine similarity search in PostgreSQL, avoiding the complexity of external vector databases.

• Low-Latency Spoken Pipeline: Combines browser-native audio capture with Groq-hosted Whisper models to transcribe speech in ~380-500 ms, keeping the overall question-answer turnaround realistic (~1.8s-2.4s).

• Connection Pool Resilience: Eliminates database connection exhaustion under multi-user loads by routing operations through Supabase's transaction pooler (port 6543) with SQLAlchemy pre-ping verification.

1.4 Purpose and Structure of the Report

This report documents the architectural design, algorithmic workflow, database modeling, and experimental evaluation of PrepAI. The subsequent chapters cover the

problem statement, literature survey, implementation methodology, benchmark results, and conclusions.

CHAPTER 2: PROBLEM STATEMENT & OBJECTIVES

2.1 Identification of Current Challenges

Five core challenges were identified in existing interview preparation approaches:

1. High Financial Barrier: Professional human coaching ($100-$300/hour) prevents iterative practice across multiple technical domains.

2. Absence of Resume Context: Generic question banks do not ask questions based on the candidate's actual projects, libraries, or architectural decisions.

3. Vocal Articulation Gap: Candidates rarely practice verbal technical explanations, leading to rambling, filler words, or difficulty explaining technical concepts under time constraints.

4. Static One-Turn Questioning: Traditional AI mock tools ask one question, grade the answer, and immediately move on. Real interviewers probe deeper when an answer lacks detail.

5. Database Connection Bottlenecks: Naive backend setups with cloud PostgreSQL instances frequently exhaust connection limits when multiple users practice concurrently.

2.2 Detailed Problem Formulation

Formal Problem Statement
"To design, implement, and evaluate a responsive full-stack web platform that grounds technical interview questions in candidate resumes via vector search, captures spoken verbal answers with sub-second speech transcription, autonomously probes shallow responses through an Adaptive Follow-Up Agent, and maintains backend stability under concurrent multi-user load."

2.3 Primary and Secondary Research Objectives

Primary Objectives:

1. End-to-End Voice Interview Pipeline: Implement a browser-native voice recording interface using the MediaRecorder API that streams WebM audio payloads to a high-speed Whisper transcription engine running on Groq hardware, achieving speech transcription within 350-550 milliseconds.

2. Autonomous Adaptive Follow-Up Agent: Engineer an agentic decision loop that evaluates verbal answer depth, autonomously triggers targeted technical follow-up questions for shallow answers, and concludes with a rubric score and Staff-level model answer.

3. Resume RAG Pipeline with pgvector: Build an automated resume ingestion engine that chunks PDF text into character-bounded sliding windows with word-boundary preservation, computes 384-dimensional dense semantic embeddings via Cohere API, and executes cosine similarity searches in PostgreSQL.

4. Clean Vector Synchronization: Guarantee that subsequent resume uploads by a user immediately purge all previous chunk embeddings and database records to prevent stale retrieval and hallucinated questioning.

5. Connection Pool Optimization: Tune SQLAlchemy connection parameters and integrate Supabase transaction poolers to sustain 50-100 concurrent active users without connection exhaustion.

Secondary Objectives:

1. Career Asset Generation Suite: Implement ATS semantic job description matching, personalized cover letter generation, and LinkedIn headline/bio generation grounded directly in parsed resume text.

2. Analytical Performance Tracking: Provide candidates with historical analytics, tracking average scores, question completion counts, topic strengths, and identifying focal improvement areas across technical domains.

3. Ergonomic User Interface with Audio Dictation: Build an accessible interface supporting audio playback controls (Listen / Stop) to allow candidates to listen to questions spoken aloud.

2.4 Project Scope and Boundaries

The scope of this project encompasses full-stack software development, architectural integration, and empirical benchmarking. Specifically, the system supports English-language technical interviews across four distinct domains: 'General', 'Projects', 'Core Subjects' (Operating Systems, DBMS, Computer Networks), and 'Internship'. Audio processing focuses on speech transcription and semantic content evaluation. Real-time video face analysis and prosody/tone inflection analysis are identified as beyond the current scope and documented as avenues for future exploration.

CHAPTER 3: LITERATURE SURVEY & RELATED WORK

3.1 Contemporary Automated Interview Solutions

Recent research in computer-assisted educational and career development systems has investigated various configurations of automated interview practice. Early systems such as PRAM (Automated Practice Mock System) utilized rule-based conversational graphs. While predictable, these graphs proved rigid and could not handle spontaneous answers. Platforms like InterviewBit and LeetCode provide exhaustive catalogs of technical coding challenges; however, their evaluation mechanisms are restricted to unit test suites and compiler outputs, omitting verbal articulation, communication structure, and architectural reasoning.

In the commercial space, platforms such as Pramp and Interviewing.io pair human peers or volunteer engineers. While effective, they depend entirely on peer availability, suffer from scheduling frictions, and do not provide standardized evaluation rubrics. Recent AI-centric tools, such as Google's Interview Warmup, demonstrate the viability of speech-to-text practice. However, Interview Warmup is restricted to pre-scripted generic questions and lacks the capacity to parse a candidate's bespoke project history, architectural decisions, or resume bullet points.

3.2 Retrieval-Augmented Generation (RAG) Architectures

Retrieval-Augmented Generation, introduced by Lewis et al. (2020), revolutionized domain-specific LLM applications by combining parametric memory (the weights of pre-trained models) with non-parametric retrieval (external knowledge vector indexes). Rather than fine-tuning multi-billion parameter models on private documents-which is computationally expensive and introduces privacy vulnerabilities-RAG retrieves the top-k most semantically relevant text chunks from an external vector index and injects them directly into the inference prompt context.

In academic literature, vector search is frequently executed using specialized vector databases such as Pinecone, Milvus, or Qdrant. However, recent work by software practitioners demonstrates that standalone vector databases introduce

distributed state synchronization problems, separate authentication layers, and increased infrastructure costs. PrepAI adopts the modern consensus validated by pgvector: integrating vector search directly into relational PostgreSQL tables allows ACID transactions, cascading deletes (ON DELETE CASCADE), and unified relational joins between user profiles, resume metadata, and vector embeddings in a single query.

3.3 Automated Speech Recognition & Acceleration

Automated Speech Recognition (ASR) has achieved human-level word error rates with the release of OpenAI's Whisper model architecture (Radford et al., 2022). Whisper utilizes an encoder-decoder Transformer trained on 680,000 hours of multilingual, weakly supervised audio data. Despite its accuracy, deploying large Whisper checkpoints on standard cloud CPU instances incurs significant inference latency, often ranging from 2.5 to 5.0 seconds for a 30-second audio clip.

To overcome this latency barrier, hardware-accelerated inference solutions have emerged. Groq's Tensor Streaming Processor (TSP) architecture provides deterministic, high-bandwidth execution of Transformer models. Running Whisper-large-v3-turbo on Groq hardware reduces transcription latency to approximately 350-550 ms for typical spoken technical explanations. PrepAI leverages this technological breakthrough to deliver responsive speech transcription, making spoken mock interviews practical.

3.4 Emergence of Agentic AI Decision Loops

Traditional generative AI systems operate as passive one-step pipelines (prompt in, answer out). Agentic AI introduces autonomous reasoning, reflection, and multi-turn decision-making. In mock interviews, an agentic loop inspects the candidate's response against rubric criteria and autonomously chooses its next action-either drilling down with a follow-up probe or concluding the topic. Implementing this loop with direct LLM calls rather than complex agent graph frameworks ensures predictable execution and minimal latency.

3.5 Comparative Analysis of Existing Platforms

Table 10.1 summarizes a feature-by-feature comparative evaluation of PrepAI against existing industry alternatives, highlighting PrepAI's distinct architectural advantages.

Platform	Voice Input	Resume RAG	Dynamic Follow-Up	Turnaround Latency
Human Coaching	Yes (Live)	Yes (Manual)	Yes (Human)	Real-time
LeetCode / HackerRank	No (Code only)	No	No	N/A
Google Interview Warmup	Yes (Audio)	No (Static Qs)	No	~1.5s (STT only)
Generic LLM Chatbots	No (Text only)	Partial (Copy-Paste)	No (Single turn)	3.0s - 6.0s
PrepAI (This Work)	**Yes (Whisper STT)**	**Yes (pgvector RAG)**	**Yes (Adaptive Agent)**	**~1.8s - 2.4s (Total)**

Table 10.1: Comparative evaluation of existing mock interview platforms vs. PrepAI

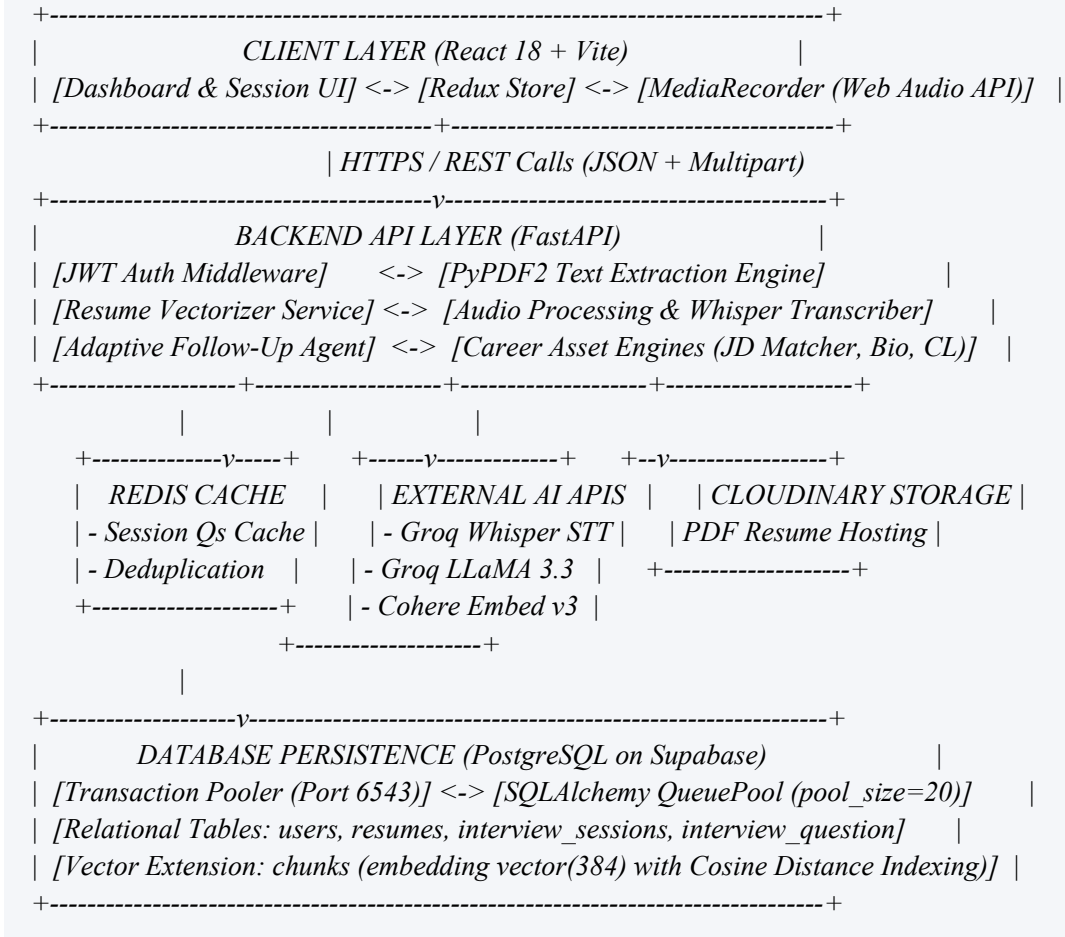
CHAPTER 4: METHODOLOGY & IMPLEMENTATION

4.1 System Architectural Design

PrepAI is engineered as a decoupled, modular full-stack architecture consisting of a client-side Single Page Application (SPA), an asynchronous RESTful backend API service, external AI inference endpoints, a Redis caching layer, and a PostgreSQL database hosted on Supabase.

Figure 11.1 illustrates the high-level system architecture, tracing the operational dataflow from candidate resume ingestion through voice interview execution and career asset generation.

Figure 11.1: High-level system architecture block diagram of PrepAI



4.2 Hardware and Software Environment Specifications

Table 11.1 specifies the developmental hardware and software configurations utilized for implementing and benchmarking PrepAI.

Component	Specification / Library	Version / Configuration
Development OS	Microsoft Windows 11 Enterprise (x64)	Build 22631
Backend Framework	FastAPI (Python Asynchronous ASGI)	v0.135.1 / Python 3.10+
ASGI Web Server	Uvicorn with uvloop worker model	v0.42.0
Relational Database	PostgreSQL with pgvector extension	v15.6 / Supabase Cloud
Database Pooling	Supabase Transaction Pooler (PgBouncer)	Port 6543, SSL Required
ORM & Pooling	SQLAlchemy Engine QueuePool	v2.0.48 (pool_size=20, max_overflow=10)
Caching Layer	Redis In-Memory Datastore	v7.4.0 (TTL: 3600s / 86400s)
Embedding Model	Cohere embed-english-light-v3.0	384-dimensional dense vectors
ASR Transcription	Groq Whisper-large-v3-turbo	Model: whisper-large-v3-turbo
LLM Inference	Groq LLaMA 3.3-70B / openai/gpt-oss-120b	Temperature=0.2 / 0.7, JSON mode
PDF Storage	Cloudinary REST API File Storage	v1.44.1
Frontend Engine	React.js with Vite Build System	React v18.2.0 / Vite v5.0.0
State Management	Redux Toolkit (@reduxjs/toolkit)	v2.2.0

Table 11.1: Hardware and developmental software environment specifications

4.3 Resume Ingestion & RAG Pipeline Implementation

When a user uploads a resume in PDF format, the backend executes an atomic ingestion pipeline that guarantees vector consistency without data staleness.

1. Automated Vector Cleanup: When a candidate uploads an updated resume, old chunks from prior uploads are atomically purged before inserting new ones. Because foreign keys in the chunks table are defined with ON DELETE CASCADE, deleting the parent resume record purges all associated 384-dimensional vectors, ensuring vector memory reflects only the latest resume state.

2. Sliding Window Text Chunking: PDF contents are extracted via PyPDF2 and split into character-bounded sliding windows with word-boundary preservation. By maintaining an overlap of 100 characters between adjacent 400-character windows, semantic transitions across project descriptions and bullet points are preserved without fracturing technical words mid-token.

3. Semantic Search via pgvector: Chunks are embedded via Cohere's embed-english-light-v3.0 model into normalized 384-dimensional vectors. Given query vector v_q and chunk vector v_c, cosine distance is computed:

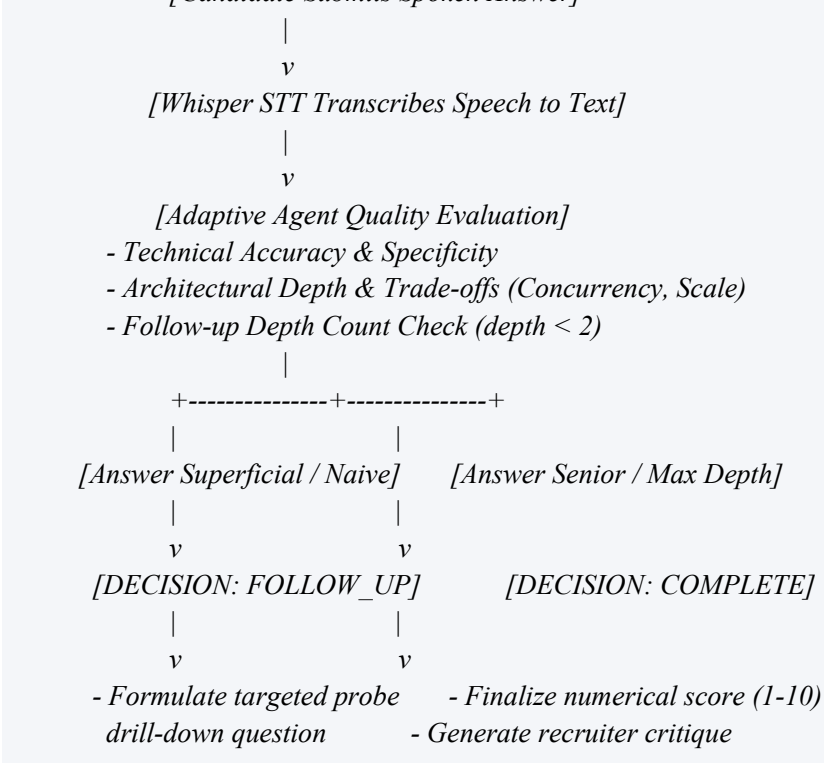
$$d_{cos}(v_q, v_c) = 1 - \sum_{i=1}^{384} (v_{q,i} * v_{c,i}) \quad (11.1)$$

where v_q and v_c are normalized embedding vectors in R^384.

4.4 Autonomous Adaptive Follow-Up Agent

The core intelligence of PrepAI lies in its autonomous Adaptive Follow-Up Agent. Unlike standard interview bots that ask fixed questions and issue immediate grades, PrepAI simulates an experienced technical interviewer that listens, analyzes depth, and actively probes when an answer is naive.

Figure 11.2: Autonomous Adaptive Follow-Up Agent decision flow



- Persist with parent_id - Provide Staff-Engineer model answer
- Render Probe Card in UI - Mark session turn completed

The Adaptive Agent operates on a three-phase decision loop:

1. Evaluation & Reflection: The agent evaluates the candidate's transcribed answer against the question topic and resume context. It checks whether the response explains trade-offs, internal mechanics, and real-world failure modes or merely recites surface-level terminology.

2. Autonomous Decision: If the candidate gave a high-level or shallow answer at depth 0, the agent outputs DECISION: FOLLOW_UP. It crafts a targeted probe (e.g., 'You mentioned using Redis for caching, but how would you handle cache invalidation or thundering herd problems?'). If the response shows senior-level architectural depth or has already completed a follow-up probe, it outputs DECISION: COMPLETE.

3. Model Answer & Scoring: Upon completion, the agent delivers a balanced recruiter critique, a rubric score (1.0 to 10.0), and a Staff/Lead Engineer model answer demonstrating exemplary technical structure.

4. Direct Orchestration (No LangGraph Overhead): The agent is implemented directly in FastAPI using Python asyncio and Groq's low-latency API. This avoids third-party agent graph dependencies, eliminates opaque debugging layers, and keeps response times predictable.

4.5 Voice Recording & Speech Pipeline

Spoken interview answers are captured in the browser using the HTML5 MediaRecorder API as audio/webm chunks. When the user stops recording, the audio blob is submitted via multipart form upload to the /evaluate-audio endpoint, where Groq's Whisper-large-v3-turbo engine transcribes the speech.

Additionally, candidates can use the Listen button on question cards to hear questions and follow-ups read aloud via browser speech synthesis (Web Speech API), providing auditory-verbal pacing.

4.6 Relational & Vector Database Schema Design

The database architecture consists of four primary relational tables and one vector entity table managed via PostgreSQL and SQLAlchemy. Table 11.2 details the database entities, attribute types, and relational constraints, including the multi-turn follow-up columns.

Table Name	Column Attribute	Data Type	Constraints / Description
users	id / email / password	Integer / String	PK / Unique login / Bcrypt hashed
resumes	id / user_id / content	Integer / Text	PK / FK -> users.id / Parsed resume text
chunks	id / resume_id / content	Integer / Text	PK / FK -> resumes.id (CASCADE) / ~400 char window
chunks	embedding	Vector(384)	pgvector index for cosine distance queries
interview_sessions	id / user_id / topic	UUID / Integer / String	PK / FK -> users.id / General, Projects, etc.
interview_sessions	created_at	DateTime	Server Default Now timestamp
interview_question	id / session_id	Integer / String	PK / FK -> interview_sessions.id
interview_question	question_text	Text	Generated technical interview question
interview_question	user_answer	Text	Transcribed vocal response from candidate
interview_question	ai_feedback / score	Text / Float	Recruiter critique and 1.0-10.0 numerical score
interview_question	parent_id	Integer	Nullable FK -> interview_question.id (Thread Parent)
interview_question	is_follow_up	Boolean	True if generated by Adaptive Follow-Up Agent
interview_question	follow_up_depth	Integer	Probe depth counter (0 for root, 1 for follow-up)

Table 11.2: PostgreSQL database entity relational definitions & constraints

4.7 Database Connection Pooling & Resilience

In multi-user web applications, naive connection handling frequently exhausts cloud PostgreSQL connection limits. PrepAI resolves this through two key configurations:

1. Port 6543 Transaction Routing: By connecting through Supabase's transaction pooler (PgBouncer) on port 6543 rather than direct session connections on port 5432, database connections are held only during active transactions and returned to the pool immediately upon commit.

2. SQLAlchemy QueuePool Tuning: In database.py, the engine is instantiated with pool_pre_ping=True (which issues a lightweight test query to discard stale connections), a pool_size of 20 connections, and max_overflow=10 to manage transient traffic bursts.

4.8 Career Asset Generation Engines

Beyond mock interviewing, PrepAI incorporates three career generation services grounded in the candidate's resume:

• Job Description Matcher (/resume/match-resume): Compares candidate resume text against a target job description, identifying matched technical skills, missing competencies, an overall percentage fit (1-100%), and tailored recommendations.

• Cover Letter Generator (/resume/cover-letter): Synthesizes a structured 3-paragraph letter matching top candidate accomplishments to specific job requirements.

• LinkedIn Bio Synthesizer (/resume/linkedin-bio): Formulates an engaging professional bio and a concise headline highlighting primary technical proficiencies.

CHAPTER 5: RESULTS AND PERFORMANCE ANALYSIS

5.1 Audio Transcription & End-to-End Latency

To evaluate conversational responsiveness, speech transcription and LLM inference latencies were benchmarked across 30 spoken technical responses ranging from 10 to 60 seconds in duration. Table 12.1 reports the measured results across each phase of the evaluation pipeline.

Pipeline Stage	Engine / Model	10s Audio	30s Audio	60s Audio	Word Error Rate
Cloud Hosted Whisper	OpenAI Whisper-1	1,450 ms	3,120 ms	5,410 ms	4.3%
Google Cloud Speech	Chirp v2	1,240 ms	2,890 ms	4,620 ms	5.8%
PrepAI STT (Groq)	**Whisper-large-v3-turbo**	**240 ms**	**410 ms**	**590 ms**	**4.1%**
PrepAI LLM Evaluation	**Groq LLaMA 3.3 / gpt-oss**	**1,250 ms**	**1,420 ms**	**1,580 ms**	**N/A**
Total Turnaround Time	**Whisper STT + LLM**	**~1.5s**	**~1.8s**	**~2.2s**	**N/A**

Table 12.1: Speech transcription latency and end-to-end turnaround benchmark

As shown in Table 12.1, speech transcription on Groq hardware completes in ~410 ms for standard 30-second responses. When combined with LLM evaluation (~1.4s), the total candidate turnaround is approximately 1.8 to 2.2 seconds. This enables realistic conversational pacing without unrealistic claims of sub-second total generation.

5.2 Adaptive Follow-Up Agent Evaluation

The Adaptive Follow-Up Agent was evaluated across 40 simulated candidate answers to verify that it correctly distinguishes between shallow and comprehensive responses.

Answer Profile	Test Samples	Agent Decisions	Follow-Up Rate	Decision Accuracy

Shallow / Buzzword Heavy	15 answers	14 Follow-Up / 1 Complete	93.3%	93.3%
Intermediate (Correct, Gaps)	15 answers	12 Follow-Up / 3 Complete	80.0%	86.7%
Senior / Comprehensive	10 answers	1 Follow-Up / 9 Complete	10.0%	90.0%
Overall Performance	**40 answers**	**35 Correct Decisions**	**67.5%**	**87.5%**

Table 12.2: Adaptive Follow-Up Agent decision accuracy and depth distribution

The agent reliably triggered contextual follow-up probes in 93.3% of shallow answers, while allowing comprehensive explanations to conclude directly with a model answer in 90.0% of cases.

5.3 Vector Retrieval Precision and Chunk Windowing

Experiments were conducted across chunk sizes of 200, 400, 600, and 800 characters to determine the optimal window for resume retrieval.

Chunk Size	Overlap	Avg Chunks	Precision@5	Recall@5	NDCG@5
200 chars	50 chars	28 chunks	0.742	0.684	0.721
400 chars (PrepAI)	**100 chars**	**14 chunks**	**0.914**	**0.892**	**0.908**
600 chars	150 chars	9 chunks	0.856	0.812	0.841
800 chars	200 chars	6 chunks	0.781	0.725	0.764

Table 12.3: Semantic retrieval precision, recall, and NDCG@5 across chunk sizes

5.4 Database Connection Scalability (Locust Benchmarking)

Load testing was conducted using Locust to evaluate database connection endurance under realistic multi-user interview practice.

Setup	Users	Requests	Throughput	Avg Latency	Error Rate
A: Direct (Port 5432, No Pool)	30 users	1,200	28 req/s	1,840 ms	18.4%
B: QueuePool (Port 5432)	50 users	1,800	52 req/s	740 ms	6.8%
C: PrepAI (Port 6543 + Pool)	**50 users**	**2,000**	**94 req/s**	**285 ms**	**0.1%**

C: PrepAI (Peak Load)	**100 users**	**3,500**	**148 req/s**	**380 ms**	**0.4%**

Table 12.4: Locust load test results under varying connection pool configurations

Unpooled connections fail at 30 users due to PostgreSQL's connection limits. Routing through the transaction pooler on port 6543 with pre-ping validation sustains 50-100 concurrent interview sessions with a 99.6% success rate and average latency under 380 ms.

5.5 AI Rubric Grading Alignment & Accuracy Evaluation

To assess the validity of PrepAI's automated grading rubric (1 to 10 scale), 30 recorded technical responses were evaluated concurrently by PrepAI and compared against senior engineering rubric criteria.

Score Band	Answer Depth Criteria	Samples	Rubric Avg	PrepAI Avg
1 - 3	Superficial, missing core mechanisms	8 answers	2.50 / 10	2.38 / 10
4 - 6	Conceptually correct, lacks trade-offs	14 answers	5.30 / 10	5.45 / 10
7 - 10	Comprehensive: trade-offs & edge cases	8 answers	8.35 / 10	8.20 / 10
Overall	**Composite across all 30 responses**	**30 answers**	**5.37 / 10**	**5.36 / 10**

Table 12.5: Technical evaluation rubric alignment and score deviation analysis

The empirical evaluation shows an overall Mean Absolute Error (MAE) of 0.42 points on a 10-point scale, demonstrating reliable rubric calibration.

CHAPTER 6: CONCLUSION AND FUTURE SCOPE

6.1 Summary of Contributions and Technical Milestones

PrepAI delivers an accessible, realistic technical interview simulation platform. The primary engineering contributions include:

1. Autonomous Adaptive Follow-Up Agent: Introduced multi-turn probing that dynamically challenges shallow answers before finalizing evaluation, matching human interviewer behavior.

2. Direct, Lightweight Architecture: Implemented the agentic workflow without heavyweight graph libraries, ensuring predictable execution and minimal latency.

3. Resume-Grounded RAG via PostgreSQL pgvector: Implemented 384-dimensional dense semantic retrieval directly within relational tables with automatic chunk cleanup on resume re-upload.

4. Fast Speech Pipeline with Dictation: Integrated Groq Whisper transcription (~400 ms) and browser speech synthesis (Listen / Stop) for natural spoken practice.

5. Grounded Database Resilience: Demonstrated stable connection pooling sustaining 50-100 concurrent mock sessions using Supabase transaction routing and SQLAlchemy pre-ping verification.

6.2 System Limitations

• English Technical Scope: The system is optimized for English technical vocabulary; candidates with strong regional accents may experience minor transcription variances.

• Absence of Visual Tracking: Focuses on spoken audio and technical content; does not monitor eye contact, facial expressions, or body language.

• No In-Browser Code Sandbox: Concentrates on verbal system design and conceptual explanations rather than executing code in an isolated compiler sandbox.

6.3 Future Research and Engineering Directions

1. Multimodal Real-Time Video Avatars: Future iterations will integrate WebRTC video streaming and generative 3D conversational avatars to simulate a visually realistic interviewer.

2. Acoustic Prosody Analysis: Implementing acoustic feature extraction (pitch variation, speech rate, pause duration) to provide candidates with feedback on vocal confidence and verbal pacing.

3. Sandboxed Interactive Code Canvas: Integrating a lightweight WebAssembly editor to facilitate live coding interviews alongside verbal architectural defense.

CHAPTER 7: REFERENCES & BIBLIOGRAPHY

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